

Erdős Problem #790

EP-790

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $l(n)$ be maximal such that if $A \subset \mathbb{Z}$ with $|A| = n$ then there exists a sum-free $B \subseteq A$ with $|B| \geq l(n)$ - that is, B is such that there are no solutions to $a_1 = a_2 + \dots + a_r$ with $a_i \in B$ all distinct.

Estimate $l(n)$. In particular, is it true that $l(n)n^{-1/2} \rightarrow \infty$? Is it true that $l(n) < n^{1-c}$ for some $c > 0$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.

Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>